

1.2.4 Data storage

Numbers

- ☐ How to convert positive denary whole numbers to binary numbers (up to and including 8 bits) and vice versa
- ☐ How to add two binary integers together (up to and including 8 bits) and explain overflow errors which may occur
- ☐ How to convert positive denary whole numbers into 2-digit hexadecimal numbers and vice versa
- ☐ How to convert binary integers to their hexadecimal equivalents and vice versa
- ☐ Binary shifts

Required

- ✓ Denary number range 0 – 255
- ✓ Hexadecimal range 00 – FF
- ✓ Binary number range 00000000 – 11111111
- ✓ Understanding of the terms 'most significant bit', and 'least significant bit'
- ✓ Conversion of any number in these ranges to another number base
- ✓ Ability to deal with binary numbers containing between 1 and 8 bits
 - e.g. 11010 is the same as 00011010
- ✓ Understand the effect of a binary shift (both left or right) on a number
- ✓ Carry out a binary shift (both left and right)

Characters

- ☐ The use of binary codes to represent characters
- ☐ The term 'character set'
- ☐ The relationship between the number of bits per character in a character set, and the number of characters which can be represented, e.g.:
 - ASCII
 - Unicode

Images

- ☐ How an image is represented as a series of pixels, represented in binary
- ☐ Metadata
- ☐ The effect of colour depth and resolution on:
 - The quality of the image
 - The size of an image file

Sound

- ☐ How sound can be sampled and stored in digital form
- ☐ The effect of sample rate, duration and bit depth on:
 - The playback quality
 - The size of a sound file

Required

- ✓ How characters are represented in binary
- ✓ How the number of characters stored is limited by the bits available
- ✓ The differences between and impact of each character set
- ✓ Understand how character sets are logically ordered, e.g. the character code for 'B' will be one more than the character code for 'A'
- ✓ Binary representation of ASCII in the exam will use 8 bits

Not required

- ✗ Memorisation of character set codes

Required

- ✓ Each pixel has a specific colour, represented by a specific code
- ✓ The effect on image size and quality when changing colour depth and resolution
- ✓ Metadata stores additional image information (e.g. height, width, etc.)

Required

- ✓ Analogue sounds must be stored in binary
- ✓ Sample rate – measured in Hertz (Hz)
- ✓ Duration – how many seconds of audio the sound file contains
- ✓ Bit depth – number of bits available to store each sample (e.g. 16-bit)